KS4 School Performance SQL Project

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1. Perspective and End Consumer

This project looks at **school performance and disadvantage** in England's secondary schools at the end of Key Stage 4 (GCSE), using 2024/25 results. It is written for a **local authority education officer**, who needs to know how the LA's schools compare with national benchmarks, which kinds of schools and pupils are falling behind, and where schools are achieving strong results despite high disadvantage.

It is a Type 1 project with 40 questions in four groups of ten. The SQL is in `KS4_Queries.sql` and the database is built by `ks4_setup.sql`.

Section	Questions	What each question does
KPI	Q1–Q10	Produces a headline figure or benchmark
Univariate	Q11–Q20	Describes the distribution of a single variable
Bivariate	Q21–Q30	Shows how one variable changes with another
Multivariate	Q31–Q40	Analyses three or more variables together

Conventions used throughout:

- A **school** is a mainstream school with a published Attainment 8 score. Special schools are excluded because most of their pupils do not sit GCSEs.
- **State-funded** means academies, free schools and LA maintained schools. Most questions use state-funded schools only, because disadvantage measures are not published for independent schools and many independent schools enter IGCSEs that do not count towards the measures.
- The **national benchmark** is the DfE's national state-funded figure; **LA benchmarks** are the DfE's local authority figures.
- **Pupil-weighted** figures weight each school by its Year 11 cohort, so they describe pupils rather than schools.
- Rankings exclude schools with fewer than 10 pupils in the cohort.

2. Datasets

Four tables are loaded into the `school_performance` database and joined on URN (school) and LA code.

Table	Source	Rows	Key columns
`ks4_results`	DfE KS4 performance tables 2024/25 (final)	5,910 (4,265 mainstream, 1,490 special, 153 LA and 2 national rows)	URN, school type, cohort size, % disadvantaged, % EAL, Attainment 8 (overall, English, maths, disadvantaged, girls, boys), % grade 4+/5+ English and maths, EBacc entry and APS, triple science, Attainment 8 in 2023 and 2024
`schools`	Get Information About Schools (GIAS)	52,616 (27,166 open)	URN, region, gender, religious character, admissions policy, trust, urban/rural classification
`la_regions`	DfE LA and region codes	153	LA code, LA name, DfE region
`school_types`	DfE abbreviations list (typed in)	27	School type code, name and group (Academy, Free school, LA maintained, Independent, Special, Alternative provision)

Loading

The KS4 file has 497 columns, so `LOAD DATA LOCAL INFILE` reads each column into a variable by position and keeps only the 31 the analysis needs; GIAS is handled the same way (26 of 135 columns). Both files are read as Latin-1 to handle Windows encoding. Suppression codes are converted to NULL and percentage signs are stripped, so every measure loads as a number.

Data quality and limitations

Issue	How the project handles it
Progress 8 is not published for 2024/25 or 2025/26, because this cohort sat no KS2 tests in 2020 and 2021	Attainment 8, English and maths passes and EBacc measures are used instead
Suppression codes (NA, NP, NE, SUPP, LOWCOV, SP) in place of numbers	Loaded as NULL and excluded from averages
Disadvantage measures are not published for independent schools	Disadvantage questions use state-funded schools only
IGCSEs and the European Baccalaureate do not count towards Attainment 8	Independent schools are kept separate; Europa School UK (0.9) is noted as a curriculum effect, not a performance one
GIAS has no Ofsted grade column, and single-word judgements ended in September 2024	Ofsted is not used
All KS4 schools matched a GIAS record and every school-type code matched the lookup	No rows lost in the joins

3. KPI Questions (Q1–Q10)

Headline figures and benchmarks that the rest of the analysis is measured against.

Q1

Question: Create a reusable view joining KS4 results to school details, type and region. Using it, how many schools and KS4 pupils are covered, in total and in state-funded schools?

Concept: VIEW over a 4-table join + COUNT / SUM KPIs

all_schools	all_ks4_pupils	state_schools	state_ks4_pupils	pct_pupils_state
4067	658433	3280	604384	91.8

1 row returned (shown in full).

Interpretation: The KS4 tables publish Attainment 8 for 4,067 mainstream schools. 3,280 of them are state-funded, and they teach 604,384 of the 658,433 pupils (91.8%). Independent schools make up about a fifth of schools but under a tenth of pupils, because their cohorts are smaller.

Q2

Question: What are the headline national figures for 2025: Attainment 8, grade 4+ and 5+ in English and maths, EBacc entry and EBacc average point score, for state-funded schools and for all schools?

Concept: Filtering the national summary rows + CASE labels

benchmark	att8	pct_em_4plus	pct_em_5plus	pct_ebacc_entry	ebacc_aps
State-funded schools	46.1	64.8	45.4	40.5	4.09
All schools	43.8	59.5	41.8	36.7	3.85

2 rows returned (shown in full).

Interpretation: In 2025, state-funded pupils averaged an Attainment 8 of 46.1. 64.8% achieved grade 4 or above in both English and maths, and 45.4% achieved grade 5 or above. 40.5% were entered for the full EBacc. Figures for all schools are lower (Attainment 8 of 43.8) because many independent schools enter IGCSEs, which do not count towards these measures.

Q3

Question: How does the national state-funded Attainment 8 compare with the simple average of school scores and the pupil-weighted average?

Concept: Benchmark subquery + simple vs weighted average

national_att8	avg_of_schools	pupil_weighted_att8
46.1	46.9	47.4

1 row returned (shown in full).

Interpretation: The average of school scores (46.9) and the pupil-weighted average (47.4) are both above the published national figure of 46.1. The difference is likely because the national state-funded figure also includes special schools and alternative provision, which this project leaves out. The pupil-weighted average is higher than the simple average, which means larger schools tend to score slightly higher.

Q4

Question: What is the national Attainment 8 gap between disadvantaged and other pupils, and the gap in the share achieving grade 5+ in English and maths?

Concept: Derived KPIs from the national summary row

pct_pupils_disadv	att8_disadv	att8_non_disadv	att8_gap	pct_em_5plus_disadv	pct_em_5plus_non_disadv	em_5plus_gap_pts
28.2	34.9	50.4	15.5	25.8	53.1	27.3

1 row returned (shown in full).

Interpretation: Nationally, 28.2% of KS4 pupils are disadvantaged. They averaged an Attainment 8 of 34.9, against 50.4 for other pupils, a gap of 15.5 points. Only 25.8% of disadvantaged pupils achieved grade 5+ in English and maths, against 53.1% of other pupils, so they are about half as likely to reach that standard.

Q5

Question: How has national state-funded Attainment 8 moved over the last three years?

Concept: Year columns in one record + derived change

att8_2023	att8_2024	att8_2025	change_1yr	change_2yr
46.3	45.9	46.1	0.2	-0.2

1 row returned (shown in full).

Interpretation: National Attainment 8 has barely moved in three years: 46.3 in 2023, 45.9 in 2024 and 46.1 in 2025. That is a 0.2-point rise on last year and a 0.2-point fall over two years, so results have settled since grading returned to pre-pandemic standards.

Q6

Question: What share of state-funded schools, and of their pupils, are above the national Attainment 8?

Concept: Conditional aggregation against a benchmark

state_schools	schools_above_national	pct_schools_above	pct_pupils_above
3280	1550	47.3	51.0

1 row returned (shown in full).

Interpretation: 1,550 of the 3,280 state-funded schools (47.3%) scored above the national Attainment 8, but they teach 51.0% of pupils. Above-average schools are therefore slightly larger on average, which matches the gap between the simple and pupil-weighted averages in Q3.

Q7

Question: What share of state-funded KS4 pupils are taught in academies and free schools rather than LA maintained schools?

Concept: GROUP BY + share of total (window SUM OVER ())

type_group	schools	ks4_pupils	pct_of_pupils
Academy	2454	463872	76.8
LA maintained	568	107996	17.9
Free school	258	32516	5.4

3 rows returned (shown in full).

Interpretation: Academies teach 76.8% of state-funded KS4 pupils and free schools a further 5.4%, so more than four in five pupils are outside LA maintained schools. LA maintained schools now teach only 17.9% of pupils, which limits the direct levers an LA education officer has over secondary results.

Q8

Question: What are the pupil-weighted rates of EBacc entry and triple science entry across state-funded schools?

Concept: Pupil-weighted averages

pct_ebacc_entry	pct_triple_sci	pct_eal
41.8	24.1	19.1

1 row returned (shown in full).

Interpretation: Weighted by pupils, 41.8% of state-funded pupils were entered for the full EBacc, 24.1% took triple science and 19.1% have English as an additional language.

Q9

Question: How many state-funded schools have fewer than a third of pupils achieving grade 5+ in English and maths, and how many pupils attend them?

Concept: Threshold KPI with COUNT / SUM and share of total

schools_below_third	pct_of_schools	pupils_affected	pct_of_pupils
778	23.7	124349	20.6

1 row returned (shown in full).

Interpretation: 778 state-funded schools (23.7%) had fewer than a third of pupils achieving grade 5+ in English and maths, and 124,349 pupils (20.6%) attend these schools. This is the group an LA would most likely prioritise for support.

Q10

Question: Create a reusable view of local authority benchmarks. Using it, how many LAs are above the national Attainment 8, and what share of pupils do they cover?

Concept: VIEW (LA rows joined to regions) + conditional KPIs

local_authorities	las_above_national	pct_las_above	pct_pupils_in_those_las
153	67	43.8	44.2

1 row returned (shown in full).

Interpretation: 67 of the 153 local authorities (43.8%) are above the national Attainment 8, and they cover 44.2% of pupils. So most LAs, and most pupils, are in areas at or below the national average.

4. Univariate Questions (Q11–Q20)

Each question describes the spread of a single variable across schools.

Q11

Question: What does the distribution of Attainment 8 look like across state-funded schools: count, mean, standard deviation, minimum and maximum?

Concept: Summary statistics (AVG, STDDEV, MIN, MAX)

schools	mean_att8	sd_att8	min_att8	max_att8
3280	46.9	9.3	0.9	87.3

1 row returned (shown in full).

Interpretation: Across 3,280 state-funded schools, Attainment 8 averages 46.9 with a standard deviation of 9.3. Scores range from 0.9 to 87.3. The minimum belongs to Europa School UK, which teaches the European Baccalaureate rather than GCSEs, so its score is not a true measure of attainment.

Q12

Question: What are the lower quartile, median and upper quartile of Attainment 8 across state-funded schools?

Concept: CTE + window functions (ROW_NUMBER, COUNT OVER) for percentiles

lower_quartile	median	upper_quartile	interquartile_range
41.0	45.7	51.1	10.1

1 row returned (shown in full).

Interpretation: The median school scores 45.7, with the middle half of schools between 41.0 and 51.1, an interquartile range of 10.1 points. The median sits below the mean (46.9), so the distribution is skewed to the right by a small number of very high-scoring schools.

Q13

Question: How many state-funded schools fall into each 5-point Attainment 8 band?

Concept: CTE + FLOOR banding (histogram) + share

att8_band	schools	pct_of_schools
0-4.9	1	0.0
20-24.9	5	0.2
25-29.9	35	1.1
30-34.9	157	4.8
35-39.9	494	15.1

att8_band	schools	pct_of_schools
40-44.9	812	24.8
45-49.9	818	24.9
50-54.9	465	14.2
55-59.9	242	7.4
60-64.9	91	2.8
65-69.9	52	1.6
70-74.9	50	1.5
75-79.9	35	1.1
80-84.9	19	0.6
85-89.9	4	0.1

15 rows returned (shown in full).

Interpretation: Half of state-funded schools (49.7%) score between 40 and 49.9. The distribution has a long upper tail reaching the 85–89.9 band, mostly selective grammar schools (see Q18). The single school in the 0–4.9 band is Europa School UK (see Q11).

Q14

Question: How is the share of disadvantaged pupils distributed across state-funded schools, in 10-point bands?

Concept: Histogram with CASE banding + share of total

disadv_band	schools	pct_of_schools
1. Under 10%	228	7.0
2. 10-19%	875	26.7
3. 20-29%	859	26.2
4. 30-39%	639	19.5
5. 40-49%	354	10.8
6. 50%+	325	9.9

6 rows returned (shown in full).

Interpretation: Most state-funded schools have between 10% and 30% disadvantaged pupils (52.9%). At the extremes, 7.0% have under 10%, while 325 schools (9.9%) have half or more of their pupils disadvantaged.

Q15

Question: How are schools with KS4 results split by school type?

Concept: Frequency table + share of total

type_name	type_group	schools	pct_of_schools
Academy (converter)	Academy	1648	40.5
Academy (sponsor led)	Academy	803	19.7
Independent school	Independent	773	19.0
Community school	LA maintained	244	6.0
Free school	Free school	196	4.8
Voluntary aided school	LA maintained	172	4.2
Foundation school	LA maintained	133	3.3
University technical college	Free school	43	1.1
Voluntary controlled school	LA maintained	19	0.5
Studio school	Free school	19	0.5
Further education institution	Other	14	0.3
City technology college	Academy	3	0.1

12 rows returned (shown in full).

Interpretation: Academies make up 60.2% of schools with KS4 results: 1,648 converters and 803 sponsor-led academies. Independent schools are 19.0%, LA maintained schools 14.0% and free schools, UTCs and studio schools 6.4%.

Q16

Question: How are state-funded schools and KS4 pupils spread across the regions of England?

Concept: Frequency table (count and sum) + share of total

region	schools	ks4_pupils	pct_of_pupils
South East	502	97034	16.1
London	498	89159	14.8
North West	460	83606	13.8
East of England	368	69356	11.5
West Midlands	385	68283	11.3
Yorkshire and the Humber	316	61296	10.1
South West	316	55479	9.2
East Midlands	286	53258	8.8
North East	149	26913	4.5

9 rows returned (shown in full).

Interpretation: The South East (16.1%) and London (14.8%) account for almost a third of state-funded KS4 pupils, while the North East has just 149 schools and 4.5% of pupils. LA comparisons in smaller regions will therefore rest on fewer schools.

Q17

Question: How large are KS4 cohorts in state-funded schools? Show the summary statistics and the share of schools in each size band.

Concept: CASE banding + aggregate statistics in one result (window functions over the grouped rows)

cohort_band	schools	pct_of_schools	smallest	largest
1. Under 100	251	7.7	13	99
2. 100-149	709	21.6	100	149
3. 150-199	1044	31.8	150	199
4. 200-249	810	24.7	200	249
5. 250+	466	14.2	250	483

5 rows returned (shown in full).

Interpretation: The most common cohort size is 150–199 pupils (31.8% of schools), and 78.1% of schools have between 100 and 249 pupils in Year 11. Cohorts range from 13 to 483 pupils.

Q18

Question: Which are the 10 highest and 10 lowest scoring state-funded schools on Attainment 8 (cohorts of 10+)?

Concept: CTE + window functions (RANK both ways)

grp	school_name	la_name	ks4_pupils	att8
Top 10	The Henrietta Barnett School	Barnet	104	87.3
Top 10	Wilson's School	Sutton	186	86.9
Top 10	Queen Elizabeth's School, Barnet	Barnet	193	85.6
Top 10	The Tiffin Girls' School	Kingston upon Thames	176	85.3
Top 10	Kendrick School	Reading	128	84.0
Top 10	St Olave's and St Saviour's Grammar School	Bromley	128	83.6
Top 10	King Edward VI Grammar School, Chelmsford	Essex	150	83.3
Top 10	King Edward VI School	Warwickshire	90	83.3
Top 10	Reading School	Reading	149	83.0
Top 10	Newstead Wood School	Bromley	156	82.8
Bottom 10	Tong Leadership Academy	Bradford	146	26.4
Bottom 10	South Shore Academy	Blackpool	134	26.3
Bottom 10	Fir Vale School	Sheffield	189	25.7
Bottom 10	South Devon UTC	Devon	66	25.7
Bottom 10	Energy Coast UTC	Cumberland	129	24.2

grp	school_name	la_name	ks4_pupils	att8
Bottom 10	Cheshire Studio School	Cheshire East	22	24.0
Bottom 10	Aylesbury UTC	Buckinghamshire	57	23.8
Bottom 10	UTC Warrington	Warrington	92	23.0
Bottom 10	Waverley Studio College	Birmingham	55	21.9
Bottom 10	Europa School UK	Oxfordshire	58	0.9

20 rows returned (shown in full).

Interpretation: All of the top 10 are selective grammar schools, led by The Henrietta Barnett School (87.3), and six are in London boroughs. Six of the bottom 10 are university technical colleges or studio schools, which recruit at age 14 and focus on technical qualifications that count less towards Attainment 8. Europa School UK's 0.9 reflects its curriculum rather than its performance.

Q19

Question: How does EBacc entry vary across state-funded schools? Group schools by the share of pupils entered.

Concept: Histogram with CASE banding, including a zero band

ebacc_entry_band	schools	pct_of_schools
1. None entered	47	1.4
2. Under 25%	1016	31.0
3. 25-49%	1107	33.8
4. 50-74%	696	21.2
5. 75%+	414	12.6

5 rows returned (shown in full).

Interpretation: Only 1.4% of state-funded schools entered no pupils for the full EBacc, but two-thirds (66.2%) entered fewer than half their pupils. Just 12.6% entered 75% or more, so full EBacc entry remains a minority route in most schools.

Q20

Question: Which state-funded schools are statistical outliers on Attainment 8, more than 2.5 standard deviations from the mean (cohorts of 10+)?

Concept: CTE + window functions (AVG and STDDEV OVER) for z-scores

school_name	la_name	ks4_pupils	att8	z_score
The Henrietta Barnett School	Barnet	104	87.3	4.34
Wilson's School	Sutton	186	86.9	4.3
Queen Elizabeth's School, Barnet	Barnet	193	85.6	4.16
The Tiffin Girls' School	Kingston upon Thames	176	85.3	4.13
Kendrick School	Reading	128	84.0	3.99
St Olave's and St Saviour's Grammar School	Bromley	128	83.6	3.94
King Edward VI Grammar School, Chelmsford	Essex	150	83.3	3.91
King Edward VI School	Warwickshire	90	83.3	3.91
Reading School	Reading	149	83.0	3.88
Newstead Wood School	Bromley	156	82.8	3.86
Pate's Grammar School	Gloucestershire	151	82.7	3.85
Colchester Royal Grammar School	Essex	128	82.1	3.78
Chelmsford County High School for Girls	Essex	180	80.7	3.63
Altrincham Grammar School for Girls	Trafford	208	80.6	3.62
Colchester County High School for Girls	Essex	191	80.6	3.62
Stratford Girls' Grammar School	Warwickshire	120	80.6	3.62

school_name	la_name	ks4_pupils	att8	z_score
Nonsuch High School for Girls	Sutton	233	80.5	3.61
UTC Warrington	Warrington	92	23.0	-2.56
Waverley Studio College	Birmingham	55	21.9	-2.68
Europa School UK	Oxfordshire	58	0.9	-4.94

Showing the 17 highest and 3 lowest of 108 rows.

Interpretation: 108 state-funded schools are more than 2.5 standard deviations from the mean, and 105 of them are above it. Almost all of the high outliers are selective grammar schools, with z-scores up to 4.34 (The Henrietta Barnett School). The notable non-selective exceptions are Michaela Community School in Brent (77.6) and Tauheedul Islam Girls' High School in Blackburn (72.3). Only three schools are low outliers: UTC Warrington, Waverley Studio College and Europa School UK.

5. Bivariate Questions (Q21–Q30)

Each question looks at how one variable changes with another.

Q21

Question: How does Attainment 8 vary by school type group?

Concept: Numeric by category: GROUP BY with AVG, MIN, MAX

type_group	schools	avg_att8	min_att8	max_att8
LA maintained	568	47.2	28.3	83.6
Academy	2454	46.9	25.7	87.3
Free school	258	45.7	0.9	77.6
Independent	773	34.5	0.2	82.9
Other	14	11.0	1.2	31.0

5 rows returned (shown in full).

Interpretation: LA maintained schools (47.2) and academies (46.9) average almost the same Attainment 8, with free schools slightly lower (45.7) because the group includes UTCs and studio schools. Independent schools average only 34.5, and further education

institutions 11.0, because their IGCSE and vocational qualifications largely do not count towards the measure, not because their pupils achieve less.

Q22

Question: How does Attainment 8 vary by region for state-funded schools?

Concept: Numeric by category + difference from national

region	schools	avg_att8	vs_national
London	498	51.1	5.0
South East	502	47.7	1.6
East of England	368	47.0	0.9
South West	316	46.7	0.6
East Midlands	286	45.6	-0.5
West Midlands	385	45.5	-0.6
North West	460	45.2	-0.9
Yorkshire and the Humber	316	45.1	-1.0
North East	149	45.0	-1.1

9 rows returned (shown in full).

Interpretation: London's state schools average 51.1, 5.0 points above the national figure and well ahead of every other region. Only four regions are above the national average, all in the South and East, while the five Midlands and northern regions sit 0.5 to 1.1 points below it, with the North East lowest at 45.0.

Q23

Question: How strongly is a school's share of disadvantaged pupils linked to its Attainment 8? Calculate the correlation across all state-funded schools.

Concept: Aggregate formula for Pearson correlation (SUM, POW, SQRT)

schools	correlation_disadv_att8
3280	-0.56

1 row returned (shown in full).

Interpretation: The correlation between a school's share of disadvantaged pupils and its Attainment 8 is −0.56, a moderately strong negative relationship. Squaring it gives about 0.31, so disadvantage alone accounts for roughly a third of the variation in school results; the rest comes from other factors, including the school itself.

Q24

Question: How does Attainment 8 differ between selective, non-selective and "non-selective in a highly selective area" state schools?

Concept: Numeric by category

admissions	schools	avg_att8
SEL	163	72.1
OTHER NON SEL	2903	45.9
NON SEL IN HIGHLY SEL AREA	214	41.5

3 rows returned (shown in full).

Interpretation: Selective schools average 72.1, 26 points above other non-selective schools (45.9). Non-selective schools in highly selective areas average just 41.5, 4.4 points lower again, because the grammar schools nearby take many of the highest attainers.

Q25

Question: Do girls', boys' and mixed state schools differ in the share of pupils achieving grade 5+ in English and maths?

Concept: Numeric by category

school_gender	schools	avg_pct_em_5plus
Boys	144	69.4
Girls	203	67.3
Mixed	2932	43.4
NULL	1	34.1

4 rows returned (shown in full).

Interpretation: Boys' schools (69.4%) and girls' schools (67.3%) have far higher shares of pupils achieving grade 5+ in English and maths than mixed schools (43.4%). This is not a like-for-like comparison, because a large share of single-sex state schools are grammar schools (see Q20). One school has no gender recorded in GIAS.

Q26

Question: Within the same schools, how much lower is Attainment 8 for disadvantaged pupils than for other pupils, and in how many schools do disadvantaged pupils do better?

Concept: Paired comparison of two columns + conditional COUNT

schools	avg_att8_disadv	avg_att8_non_disadv	avg_gap	schools_no_gap
3249	37.8	50.0	12.2	38

1 row returned (shown in full).

Interpretation: Across 3,249 schools with both figures, disadvantaged pupils average an Attainment 8 of 37.8 against 50.0 for their classmates, a within-school gap of 12.2 points. This is smaller than the national gap of 15.5 (Q4), because part of the national gap comes from disadvantaged pupils attending lower-attaining schools. Only 38 schools have no gap at all.

Q27

Question: In mixed state schools, how do girls' and boys' Attainment 8 compare, and in what share of schools do girls outperform boys?

Concept: Paired comparison of two columns + share

mixed_schools	avg_att8_girls	avg_att8_boys	avg_gap	pct_girls_ahead
2913	46.9	44.0	2.9	80.2

1 row returned (shown in full).

Interpretation: In 2,913 mixed state schools, girls average an Attainment 8 of 46.9 against 44.0 for boys, a gap of 2.9 points. Girls outperform boys in 80.2% of mixed schools, so the gender gap is consistent rather than driven by a few schools.

Q28

Question: Is higher EBacc entry associated with higher Attainment 8? Compare average Attainment 8 across EBacc entry bands.

Concept: CASE banding of one variable + AVG of another

ebacc_entry_band	schools	avg_att8
1. Under 25%	1063	41.7
2. 25-49%	1107	45.8
3. 50-74%	696	50.1
4. 75%+	414	57.5

4 rows returned (shown in full).

Interpretation: Average Attainment 8 rises steadily with EBacc entry, from 41.7 in schools entering under a quarter of pupils to 57.5 in schools entering three-quarters or more. The relationship runs both ways: higher-attaining schools enter more pupils, and the EBacc subjects carry full weight in Attainment 8, so this should not be read as EBacc entry alone raising results.

Q29

Question: How did state schools' Attainment 8 in 2025 compare with their own 2024 score? How many improved, held steady (within 1 point) or declined?

Concept: Paired comparison of two years + CASE grouping

direction	schools	pct_of_schools	avg_change
1. Improved (1+ point)	1139	35.3	2.9
2. Stable (within 1 point)	1043	32.3	0.0
3. Declined (1+ point)	1049	32.5	-2.8

3 rows returned (shown in full).

Interpretation: Although national Attainment 8 rose by only 0.2 points (Q5), individual schools moved much more. 35.3% of schools improved by more than a point (by 2.9 on average), and 32.5% declined by a similar amount (2.8). Only 32.3% stayed within a point of their 2024 score, so year-to-year change at school level is common.

Q30

Question: How does Attainment 8 differ between urban, larger rural and smaller rural state schools?

Concept: Numeric by a derived (string function) category

urban_rural	schools	avg_att8
Urban	2830	47.1
Larger rural	297	46.0
Smaller rural	119	45.8
(England/Wales) Urban major conurbation	17	44.6
(England/Wales) Urban city and town	14	41.9
(England/Wales) Rural town and fringe	3	40.9

6 rows returned (shown in full).

Interpretation: Urban state schools average 47.1, slightly ahead of larger rural (46.0) and smaller rural schools (45.8), so the urban–rural gap is about one point. 34 schools still carry older 2011 classifications in GIAS, shown with the "(England/Wales)" prefix; they are too few to change the overall picture but would need mapping to the current categories in a fuller analysis.

6. Multivariate Questions (Q31–Q40)

Each question analyses three or more variables together.

Q31

Question: Split state schools into five equal groups by their share of disadvantaged pupils. How do Attainment 8, strong English and maths passes and EBacc entry change from the least to the most disadvantaged fifth?

Concept: CTE + window function (NTILE) + GROUP BY

disadv_quintile	schools	min_pct_disadv	max_pct_disadv	avg_att8	avg_pct_em_5plus	avg_pct_ebacc_entry
1	656	0.0	15.7	56.8	65.2	53.3
2	656	15.8	22.2	48.4	49.1	41.8
3	656	22.2	30.0	45.4	43.1	37.5
4	656	30.0	40.4	42.9	38.2	35.6
5	656	40.5	80.1	40.9	34.4	35.8

5 rows returned (shown in full).

Interpretation: Attainment 8 falls at every step from the least to the most disadvantaged fifth of schools, from 56.8 to 40.9, and the share achieving grade 5+ in English and maths almost halves from 65.2% to 34.4%. The biggest drop is between the first and second fifths (8.4 points), so the most advantaged schools stand furthest apart. EBacc entry levels off at about 36% in the two most disadvantaged fifths.

Q32

Question: Do state schools with a religious character perform differently once their level of disadvantage is taken into account? Compare faith and non-faith schools within low, medium and high disadvantage bands.

Concept: CASE banding + two-way GROUP BY

faith_status	disadv_band	schools	avg_att8	avg_att8_disadv
Faith school	1. Low (<20%)	219	55.3	43.8
No religious character	1. Low (<20%)	884	53.1	40.9
Faith school	2. Medium (20-40%)	261	46.3	37.6
No religious character	2. Medium (20-40%)	1237	44.3	35.5
Faith school	3. High (40%+)	114	42.8	37.8
No religious character	3. High (40%+)	565	40.6	35.9

6 rows returned (shown in full).

Interpretation: Faith schools score about 2 points higher than schools with no religious character in every disadvantage band (55.3 v 53.1, 46.3 v 44.3, 42.8 v 40.6), and their disadvantaged pupils also do better by 1.9 to 2.9 points. The advantage holds once disadvantage is controlled for, though faith admissions criteria may still shape intakes in ways these bands cannot capture.

Q33

Question: In each region, how large is the gap in Attainment 8 between disadvantaged and other pupils, and in the share achieving grade 5+ in English and maths?

Concept: GROUP BY + two derived gap measures

region	schools	avg_pct_disadv	att8_disadv	att8_non_disadv	att8_gap	em_5plus_gap_pts
London	492	34.9	44.3	54.0	9.7	19.2
West Midlands	379	33.6	37.5	48.6	11.0	20.5

region	schools	avg_pct_disadv	att8_disadv	att8_non_disadv	att8_gap	em_5plus_gap_pts
North West	454	31.3	37.0	48.7	11.6	21.1
East Midlands	285	26.6	36.6	48.7	12.2	22.8
North East	149	35.2	36.5	49.4	12.8	24.5
Yorkshire and the Humber	313	30.5	35.6	48.8	13.2	24.0
South East	497	21.8	37.0	50.4	13.4	23.5
East of England	367	22.6	36.2	49.8	13.5	24.7
South West	313	22.7	35.9	49.5	13.6	24.4

9 rows returned (shown in full).

Interpretation: London has both the highest share of disadvantaged pupils (34.9%) and the smallest gap: its disadvantaged pupils average 44.3, far above any other region, and trail their classmates by 9.7 points. The largest gaps are in the South West (13.6), East of England (13.5) and South East (13.4), the three least disadvantaged regions, where disadvantaged pupils are a smaller minority and fall further behind.

Q34

Question: Which state schools are "beating the odds": in the top quarter for disadvantage, yet with Attainment 8 above the national state-funded average? Show the 20 highest scoring.

Concept: CTE + window function (PERCENT_RANK) + benchmark subquery

school_name	la_name	type_name	ks4_pupils	pct_disadv	att8	national_att8
Mossbourne Community Academy	Hackney	Academy (sponsor led)	206	52.4	65.1	46.1
Mossbourne Victoria Park Academy	Hackney	Academy (sponsor led)	156	58.3	64.3	46.1
The Camden School for Girls	Camden	Voluntary aided school	117	38.5	64.0	46.1
Eden Girls' School, Waltham Forest	Waltham Forest	Free school	122	40.2	63.5	46.1
Eden Girls' Leadership Academy, Birmingham	Birmingham	Free school	120	40.8	62.1	46.1
The St Thomas the Apostle College	Southwark	Voluntary aided school	161	45.3	61.7	46.1

school_name	la_name	type_name	ks4_pupils	pct_disadv	att8	national_att8
Paddington Academy	Westminster	Academy (sponsor led)	185	51.4	61.6	46.1
Eden Boys' School, Birmingham	Birmingham	Free school	122	51.6	61.3	46.1
Eden Boys' Leadership Academy, Birmingham East	Birmingham	Free school	123	48.0	60.8	46.1
Brampton Manor Academy	Newham	Academy (converter)	361	47.1	59.9	46.1
Clapton Girls' Academy	Hackney	Academy (converter)	175	45.1	59.4	46.1
The Hurlingham Academy	Hammersmith and Fulham	Academy (sponsor led)	147	39.5	58.4	46.1
Harris Academy Wimbledon	Merton	Free school	171	39.8	58.3	46.1
Eden Girls' Leadership Academy, Manchester	Manchester	Free school	129	55.0	58.2	46.1
Forest Gate Community School	Newham	Academy (converter)	264	45.5	58.1	46.1
St Aloysius RC College	Islington	Voluntary aided school	63	47.6	57.9	46.1
Highbury Fields School	Islington	Community school	134	53.7	57.4	46.1
Fulham Cross Girls' School and Language College	Hammersmith and Fulham	Academy (converter)	111	39.6	57.1	46.1
Didsbury High School	Manchester	Free school	218	43.1	57.1	46.1
Parliament Hill School	Camden	Community school	179	40.2	57.1	46.1

20 rows returned (shown in full).

Interpretation: All 20 schools have at least 38% disadvantaged pupils yet score well above the national 46.1, led by Mossbourne Community Academy in Hackney (65.1 with 52.4% disadvantaged). 15 of the 20 are in London and five are in Birmingham and Manchester. Star Academies' Eden schools account for five of the 20, four outside London and one in Waltham Forest. High disadvantage clearly does not rule out high attainment.

Q35

Question: How do the share of pupils with English as an additional language and disadvantage combine? Compare Attainment 8 for schools grouped by both.

Concept: Two CASE bandings + cross-tab with conditional aggregation

eal_band	schools	att8_low_disadv	att8_mid_disadv	att8_high_disadv
1. Low EAL (<10%)	1451	51.1	42.8	36.8
2. Medium EAL (10-30%)	1122	56.1	45.1	39.3
3. High EAL (30%+)	707	61.9	48.0	43.6

3 rows returned (shown in full).

Interpretation: Within every disadvantage band, schools with more pupils who have English as an additional language score higher. Among the most disadvantaged schools, Attainment 8 rises from 36.8 where EAL is low to 43.6 where it is high. EAL is therefore not a barrier to attainment at KS4, and the lowest results are in high-disadvantage schools with few EAL pupils.

Q36

Question: Is the link between disadvantage and attainment stronger in some regions than others? Calculate the correlation within each region alongside its average disadvantage and Attainment 8.

Concept: Pearson correlation formula + GROUP BY

region	schools	avg_pct_disadv	avg_att8	correlation
South East	502	21.6	47.7	-0.74
North East	149	35.2	45.0	-0.72
East Midlands	286	26.5	45.6	-0.71
East of England	368	22.5	47.0	-0.7
Yorkshire and the Humber	316	30.3	45.1	-0.69
South West	316	22.6	46.7	-0.66
North West	460	31.1	45.2	-0.64
London	498	34.6	51.1	-0.58

region	schools	avg_pct_disadv	avg_att8	correlation
West Midlands	385	33.2	45.5	-0.47

9 rows returned (shown in full).

Interpretation: The link between disadvantage and attainment is strongest in the South East (correlation −0.74), North East (−0.72) and East Midlands (−0.71), and weakest in London (−0.58) and the West Midlands (−0.47). In London and the West Midlands, a school's intake predicts its results less well, partly because high-performing schools there serve disadvantaged communities.

Q37

Question: Using the LA benchmarks, which local authorities have above-average disadvantage but still beat the national Attainment 8, and how do their disadvantaged pupils do?

Concept: Uses v_la_benchmarks + CROSS JOIN to the national row

la_name	dfe_region	pct_disadv	national_pct_disadv	att8	national_att8	att8_vs_national	att8_disadv
Hammersmith and Fulham	London West	33.2	28.2	55.5	46.1	9.4	45.2
Kensington and Chelsea	London Central	35.8	28.2	54.5	46.1	8.4	46.0
Westminster	London Central	43.2	28.2	53.2	46.1	7.1	45.9
Merton	London South	33.6	28.2	52.3	46.1	6.2	45.3
Southwark	London Central	45.1	28.2	52.1	46.1	6.0	44.9
Ealing	London West	31.8	28.2	52.0	46.1	5.9	44.5
Wandsworth	London Central	37.3	28.2	51.8	46.1	5.7	41.2
Brent	London West	28.8	28.2	51.7	46.1	5.6	42.8
Newham	London East	51.1	28.2	51.2	46.1	5.1	48.4
Hackney	London East	53.8	28.2	50.5	46.1	4.4	45.0
Hounslow	London West	28.8	28.2	49.9	46.1	3.8	42.4
Haringey	London North	37.1	28.2	48.5	46.1	2.4	40.3
Barking and Dagenham	London East	32.1	28.2	48.4	46.1	2.3	41.6

la_name	dfe_region	pct_disadv	national_pct_disadv	att8	national_att8	att8_vs_national	att8_disadv
Tower Hamlets	London East	55.4	28.2	47.4	46.1	1.3	45.7
Islington	London Central	56.0	28.2	47.0	46.1	0.9	40.9
Solihull	West Midlands C	28.6	28.2	47.0	46.1	0.9	33.4
Gateshead	North East B	30.9	28.2	47.0	46.1	0.9	33.9
Camden	London Central	55.4	28.2	46.9	46.1	0.8	40.6
Greenwich	London East	32.4	28.2	46.9	46.1	0.8	39.1
Lewisham	London East	37.7	28.2	46.7	46.1	0.6	37.9
Birmingham	West Midlands C	48.2	28.2	46.6	46.1	0.5	40.2
Kirklees	South and West Yorkshire	31.0	28.2	46.4	46.1	0.3	35.6
Bristol, City of	South West B	32.8	28.2	46.4	46.1	0.3	33.8
Enfield	London North	34.3	28.2	46.3	46.1	0.2	37.5

24 rows returned (shown in full).

Interpretation: 24 local authorities have more disadvantaged pupils than the national average and still beat the national Attainment 8, and 19 of them are London boroughs. Hammersmith and Fulham leads at 9.4 points above national. In Newham, where 51.1% of pupils are disadvantaged, disadvantaged pupils average 48.4, above the national figure for all pupils. The five LAs outside London (Solihull, Gateshead, Birmingham, Kirklees and Bristol) only just beat national, and their disadvantaged pupils score 33.4–40.2.

Q38

Question: Within each region, how do academies and LA maintained schools compare on disadvantage and Attainment 8?

Concept: Two-way GROUP BY (region x type group) + HAVING

region	type_group	schools	avg_pct_disadv	avg_att8
East Midlands	Academy	253	26.5	45.5
East Midlands	LA maintained	21	26.1	47.5
East of England	Academy	316	22.2	47.1
East of England	LA maintained	23	22.5	46.6

region	type_group	schools	avg_pct_disadv	avg_att8
London	Academy	305	33.4	51.2
London	LA maintained	127	36.0	50.8
North East	Academy	122	35.1	45.3
North East	LA maintained	20	33.2	44.4
North West	Academy	291	31.9	45.3
North West	LA maintained	135	27.9	45.4
South East	Academy	346	21.5	48.2
South East	LA maintained	116	21.1	47.8
South West	Academy	263	22.1	47.5
South West	LA maintained	33	22.7	44.9
West Midlands	Academy	316	32.2	45.7
West Midlands	LA maintained	43	34.9	45.0
Yorkshire and the Humber	Academy	242	30.9	44.8
Yorkshire and the Humber	LA maintained	50	25.8	46.0

18 rows returned (shown in full).

Interpretation: Within each region, academies and LA maintained schools have similar disadvantage levels and differ by no more than 2.6 points in Attainment 8. Academies are ahead in six regions and LA maintained schools in three (East Midlands, North West and Yorkshire and the Humber), so school type explains little once region is held constant.

Q39

Question: Among multi-academy trusts running five or more secondary schools in the KS4 tables, which have the highest and lowest average Attainment 8, and how disadvantaged are their intakes?

Concept: CTE + GROUP BY/HAVING + window functions (RANK both ways)

grp	trust_name	schools	ks4_pupils	avg_pct_disadv	avg_att8	avg_att8_disadv
Top 10	KING EDWARD VI ACADEMY TRUST BIRMINGHAM	12	1859	37.9	61.4	57.5

grp	trust_name	schools	ks4_pupils	avg_pct_disadv	avg_att8	avg_att8_disadv
Top 10	RUSSELL EDUCATION TRUST	5	790	18.8	54.4	38.0
Top 10	THE DIOCESE OF WESTMINSTER ACADEMY TRUST	6	1181	18.4	54.2	45.1
Top 10	THE LAURUS TRUST	6	1413	24.6	54.2	43.5
Top 10	WEYDON MULTI ACADEMY TRUST	6	1366	13.1	53.4	35.6
Top 10	IMPACT MULTI ACADEMY TRUST	5	1183	11.8	52.9	40.2
Top 10	LEARNING PARTNERS ACADEMY TRUST	6	1084	13.4	51.9	33.6
Top 10	BISHOP WILKINSON CATHOLIC EDUCATION TRUST	5	1178	19.3	51.3	40.4
Top 10	STAR ACADEMIES	23	3103	40.3	51.1	47.8
Top 10	EASTERN LEARNING ALLIANCE	5	854	21.8	50.7	37.4
Bottom 10	ACADEMY TRANSFORMATION TRUST	10	1520	34.8	39.8	31.3
Bottom 10	BRADFORD DIOCESAN ACADEMIES TRUST	5	1090	39.4	39.8	32.6
Bottom 10	CONSILIUM ACADEMIES	7	995	38.1	39.4	31.7
Bottom 10	SWALE ACADEMIES TRUST	9	1729	32.4	38.8	29.6
Bottom 10	THE CO-OPERATIVE ACADEMIES TRUST	12	2800	48.8	38.4	32.9
Bottom 10	ACTIVATE LEARNING EDUCATION TRUST	6	581	26.7	37.6	28.0
Bottom 10	THE KEMNAL ACADEMIES TRUST	14	2117	32.9	37.5	30.0
Bottom 10	FYLDE COAST ACADEMY TRUST	5	794	46.0	37.3	30.1
Bottom 10	WICKERSLEY PARTNERSHIP TRUST	5	941	46.1	36.8	30.3
Bottom 10	THE LEARNING PARTNERSHIP ACADEMIES TRUST	6	741	33.3	36.7	30.1

grp	trust_name	schools	ks4_pupils	avg_pct_disadv	avg_att8	avg_att8_disadv
Bottom 10	ALDRIDGE EDUCATION	8	1172	43.6	36.2	30.8

21 rows returned (shown in full).

Interpretation: King Edward VI Academy Trust Birmingham leads with an average of 61.4, largely because its schools include grammar schools, though 37.9% of its pupils are disadvantaged. Star Academies stands out: across 23 schools with 40.3% disadvantaged pupils, it averages 51.1, and its disadvantaged pupils score 47.8, above the national average for all pupils. The bottom trusts average 36–40, mostly with high disadvantage. The bottom group has 11 trusts because two tie at 39.8.

Q40

Question: In each local authority, which state school has the highest Attainment 8, and how do its disadvantage rate and English and maths results compare with the LA's lowest scoring school?

Concept: CTE + window functions (ROW_NUMBER both ways, COUNT OVER) + self-join of the top and bottom rows

la_name	region	la_schools	top_school	top_pct_disadv	top_att8	top_pct_em_5plus	bottom_school	bottom_pct_disadv	bottom_att8	bottom_pct_em_5plus
Derby	East Midlands	16	Landau Forte College	35.9	55.4	65.1	Alvaston Moor Academy	67.7	32.4	23.4
Derbyshire	East Midlands	45	St Mary's Catholic High School, A Catholic Voluntary Academy	15.0	59.4	73.3	Mercia Academy	35.7	32.6	23.2
Leicester	East Midlands	20	Madani Girls' School	21.3	61.4	73.4	Lancaster Academy	39.4	37.5	26.1
Leicestershire	East Midlands	43	The Market Bosworth School	5.2	59.5	72.1	The Winstanley School	43.2	29.9	17.8
Lincolnshire	East Midlands	54	The King's School, Grantham	6.2	72.7	94.9	William Lovell Church of England Academy	50.9	29.4	12.7
North Northamptonshire	East Midlands	20	Bishop Stopford School	7.9	58.7	66.0	Lodge Park Academy	34.2	33.5	20.9
Nottingham	East Midlands	18	The Trinity Catholic School A Voluntary Academy	16.3	55.3	59.2	Bluecoat Beechdale Academy	70.5	34.0	21.4
Nottinghamshire	East Midlands	45	The West Bridgford School	5.6	62.9	80.7	Queen Elizabeth's Academy	49.2	32.2	17.8

la_name	region	la_schools	top_school	top_pct_disadv	top_att8	top_pct_em_5plus	bottom_school	bottom_pct_disadv	bottom_att8	bottom_pct_em_5plus
West Northamptonshire	East Midlands	22	Northampton School for Boys	14.2	58.2	71.6	Silverstone UTC	20.5	31.1	14.5
Bedford	East of England	13	Goldington Academy	25.4	54.3	64.4	Daubeney Academy	23.9	38.5	29.4
Cambridgeshire	East of England	34	Chesterton Community College	13.7	68.0	83.7	Neale-Wade Academy	35.3	34.8	23.7
Central Bedfordshire	East of England	15	Cedars Upper School	14.5	48.5	49.5	All Saints Academy Dunstable	33.6	36.9	25.2
Essex	East of England	79	King Edward VI Grammar School, Chelmsford	2.7	83.3	100.0	Bmat Stem Academy	37.6	28.1	15.3
Hertfordshire	East of England	79	Dame Alice Owen's School	9.5	72.9	87.5	The Astley Cooper School	32.9	34.7	26.7
Luton	East of England	13	Denbigh High School	27.4	52.6	56.2	The Stockwood Park Academy	43.5	35.9	20.7
Norfolk	East of England	53	Notre Dame High School, Norwich	13.0	61.3	69.3	City Academy Norwich	50.4	35.2	29.6
Peterborough	East of England	14	The King's (The Cathedral) School	11.6	63.2	75.5	Queen Katharine Academy	50.0	32.5	22.6
Southend-on-Sea	East of England	12	Westcliff High School for Boys Academy	9.6	77.6	97.7	Cecil Jones Academy	42.6	35.1	21.3
Suffolk	East of England	43	Debenham High School	10.1	55.7	68.1	Beccles High School	43.3	27.6	17.9
Thurrock	East of England	13	Grays Convent High School	16.1	55.7	61.3	Ormiston Park Academy	39.4	37.5	28.3

Showing 20 of 150 rows.

Interpretation: In the 150 LAs with five or more schools, the best school outscores the weakest by 28.5 points on average. In 146 of them the best school also has a smaller share of disadvantaged pupils, by 27.2 points on average. At least 18 of the top schools are grammar schools and 16 of the bottom are UTCs or studio schools, which widens the gap. Oxfordshire's 60.4-point range is inflated by Europa School UK; the narrowest is Knowsley at 11.4. The first 20 rows are shown; the full output covers every region.

7. Summary of Findings

Disadvantage is the strongest single influence on school results, but where a school is and how it is run can partly offset it.

Section	Key finding
KPI	Disadvantaged pupils average 34.9 in Attainment 8 against 50.4 for other pupils, and are half as likely to achieve grade 5+ in English and maths (25.8% v 53.1%) (Q4). National results have been flat for three years (Q5), and 778 schools have fewer than a third of pupils reaching grade 5+ in English and maths (Q9).
Univariate	Most schools score between 40 and 50, with a long upper tail of grammar schools (Q13, Q20). The lowest scores mostly belong to UTCs, studio schools and Europa School UK, which reflects their curricula more than their teaching (Q18).
Bivariate	Disadvantage correlates with Attainment 8 at −0.56 (Q23). London leads every region by at least 3.4 points (Q22). Non-selective schools in grammar areas score lowest of all admissions types (Q24), and girls outperform boys in four out of five mixed schools (Q27).
Multivariate	London's disadvantaged pupils average 44.3 and face the smallest gap (Q33). 19 of the 24 high-disadvantage LAs that beat national are London boroughs (Q37). Star Academies, Mossbourne and other "beating the odds" schools show high disadvantage does not rule out high attainment (Q34, Q39). Higher EAL is linked to higher results at every level of disadvantage (Q35), while academy status makes little difference once region is held constant (Q38).

For an LA education officer, the priorities are the high-disadvantage, low-EAL schools outside London, where the disadvantage gap is widest. The "beating the odds" schools in Q34 and Q37 are the obvious places to learn from.

The main limitation is the lack of Progress 8 this year. Attainment 8 measures where pupils end up, not how much progress they made, so the school-level comparisons reflect intake as well as teaching.